



Jordan University of Science & Technology
Department of Applied Biological Sciences
General Microbiology (BT231)
Second Semester 2022-2023

Course instructor: Professor Homa Darmani, Professor Ziad Jaradat
Course Coordinator: Professor Homa Darmani

Text book; Prescott's Microbiology by Willey, Sherwood and Woolverton, 10th edition 2017, McGraw- Hill Publisher

Course description: This course will give an introduction to the microbial world. Students will gain knowledge of the diversity of prokaryotes, their development, structure and function. They will gain an understanding of prokaryotic metabolism, nutrition, growth and different methods that are used to control microbial growth in industrial and medical settings, as well as, the use of antimicrobial chemotherapeutic drugs to treat infections. The major classes of bacteria, viruses and fungi will be thoroughly discussed and students will gain knowledge about host-pathogen relationships.

Course outcomes: Upon completion of the course, the students should be able to:

Outcome	Weight
Describe: the surface structure and internal structure of bacterial cells and viruses and their functions; other acellular agents of importance; the nutritional and physical requirements for bacterial growth.	30%
Describe: the effect of the environment on bacteria and the dynamics and measurement of the growth of a bacterial population; the use of physical and chemical methods including antimicrobial chemotherapeutic agents to control microbial growth in industrial and medical settings; microbial taxonomy using molecular methods.	30%
Be familiar with: the major groups of bacteria and fungi and their importance in the environment, in medicine, and in the food industry; the factors that influence transmission of pathogens and the occurrence of infectious diseases.	40%

Topic	Chapter	Pages
1. Bacterial Cell Structure	3	42-77
2. Viruses and other acellular agents	6	109-130
3. Microbial Growth	7	132-169
4. Control of microorganisms in the environment	8	172-186
5. Antimicrobial chemotherapy	9	188-206
6. Microbial Taxonomy and evolution of diversity	19	443-452
7. Bacteria: The Proteobacteria	22	504-537
8. Firmicutes: The low G+C gram positive bacteria	23	539-550
9. Actinobacteria: The high G+C gram positive bacteria	24	568-580
10. Fungi (Eumycota)	26	583-596
11. Pathogenicity and infection	35	770-784

Assessment		
Assessment Type	Expected Due Date	Weight
First Exam	To be determined (TBD)	22%
Second Exam	TBD	22%
Quizzes	Every Lecture	16%
Final Exam	TBD	40%
Total		100%

Course Policies

1. **Class attendance is mandatory.** Absences in excess of 20% of the total lecture hours will result in your being dropped from the course with a failing grade.
2. **Make-up exam** appeals should be filed within **Two days** of the missed exam.
3. **Quizzes:** The grade for each quiz will be given to students who answer the question correctly AND who scan the QR code –**a student who does not scan the QR CODE for attendance (for whatever reason) will lose the mark.**
4. **Cell phones are prohibited** during examinations and must be turned off during the lecture. No incoming or outgoing calls or text messages are allowed.
5. **Unethical conduct, including cheating** during examinations, will result in punishment by the university administration as per cheating rules and regulations.